

4.9 Creation of Tables in Microsoft Access

پہلے ٹیبل کا منصوبہ بنائیں۔

CS XI

Chapter 4: Data and Analysis

Lecture 11 of 15

4.9 Creation of Tables in Microsoft Access

LEARN IT

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

Datasheet View (4.9.2) — fast, spreadsheet-like, good for beginners, weak on rules.

1. Create tab → Table.
2. A blank grid appears.
3. Double-click Field1, Field2, ... to rename.
4. Type data; Access guesses the data type.

Limitation: you cannot easily say "Marks cannot exceed 100". Professionals use Design View.

Design View (4.9.3) — full control before any data is typed.

1. Create tab → Table Design.
2. Type each Field Name (StudentID, Name, ...).
3. Pick the Data Type from the dropdown.
4. Right-click the unique field → Primary Key (key icon).
5. Save the table.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

COMMON MISTAKE

Phone numbers look numeric but are Short Text. Leading zeros (0333...) would vanish in a Number field, and you never sum two phone numbers.

اردو میں سمجھیں

پہلے ٹیبل کا منصوبہ بنائیں۔ ڈیٹا شیٹ ویو تیز لیکن کمزور ہے؛ ڈیزائن ویو میں فیلڈ، ڈیٹا ٹائپ اور پرائمری کی خود منتخب کرتے ہیں۔ فون نمبر شارٹ ٹیکسٹ رکھیں۔

CHECK YOURSELF

1. Best data type for StudentID if you want Access to generate it?
2. Best data type for FeeAmount?

Answers: 1. AutoNumber. 2. Currency.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Design View is best when you:

- a. Need full control of names, types, keys before data
- b. Want a random spreadsheet
- c. Draw Figma
- d. Ping a router

MCQ 2. Datasheet View is for:

- a. Typing rows after the structure exists
- b. Defining PK only
- c. Writing Python
- d. OSI captures

MCQ 3. A Number field should not store:

- a. Phone numbers that need leading zeros (use Text)
- b. Marks
- c. Counts
- d. IDs that are truly numeric

MCQ 4. Primary key is set in:

- a. Design View (key icon)
- b. A report footer
- c. A pie
- d. Agile sticky

MCQ 5. Yes/No fields store:

- a. Boolean-like presence
- b. Paragraphs
- c. Images only
- d. OSI layers

MCQ 6. Creating tables is step one because:

- a. Everything else hangs on the structure
- b. Reports can store data
- c. Queries invent columns from nothing always
- d. Forms replace keys

Extra short questions

1. Steps to create a table in Design View.
2. When Datasheet create is OK?
3. Why set field size/validation?
4. Name three data types you used in class.

Extra long question

1. Create (on paper) Design View for Students: four fields, PK, one validation rule, and say what error Access shows if the rule fails.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Create → Table Design; field name; data type; PK; save name.
2. Quick throwaway; still go back to Design to set PK/types.
3. Stop 999 marks out of 100; stop 40-character roll numbers.
4. Short Text, Number, Date/Time, Yes/No, Currency, AutoNumber.

Long

1. Roll AutoNumber/Number PK; Name Short Text required; Marks Number 0–100 validation; Passed Yes/No. Invalid marks: validation text "0 to 100".

Next lecture: 4.10 Designing Forms for Data Manipulation